

Genetic Algorithm

CSC3034 Computational Intelligence

Dr Tang Tiong Yew

Fundamental ideas of genetic algorithm

Structure of a genetic algorithm

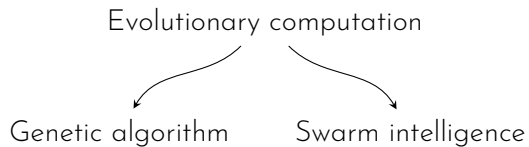
Selection

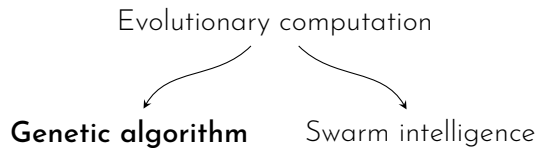
Problem formulation

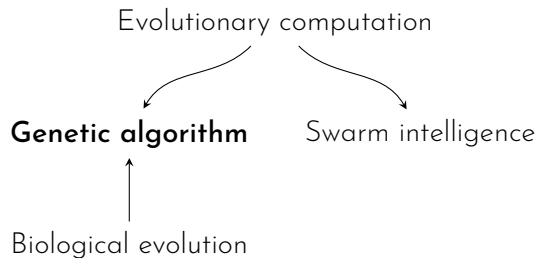
Example

Evolution

Variants







Biological evolution

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an optimisation process to improve the ability of an organism (or system) to survive in dynamically changing and competitive environments.

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Charles Darwin developed the theory of **natural selection** which became the foundation of biological evolution.

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- ▶ Over time, these good characteristics become dominant among the population.

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What if an organism mutated and developed worse characteristics?

These processes (**natural selection**, **survival of the fittest**, **reproduction**, and **mutation**) form the fundamental components for the genetic algorithm.

- ▶ Each member in a population is considered as an individual **chromosome**.

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- ▶ The characteristics of an individual are encoded as **genes** in the chromosome.

Generic genetic algorithm

Genetic algorithm is

an optimisation process of which the purpose is to identify the chromosome with the best combination of genes.

Optimisation problem

What is an optimisation problem?

SUNWAY
UNIVERSITY

Jeffrey Cheah
Foundation

Nurturing the Seeds of Wisdom

- ▶ finding the value that will maximise/minimise another value(s)

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- ▶ are search problems optimisation problems?

Generic genetic algorithm

Genetic algorithm is an algorithm to solve an optimisation problem in which we are

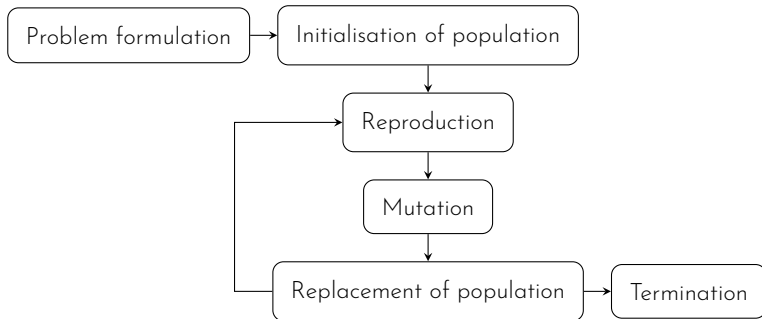
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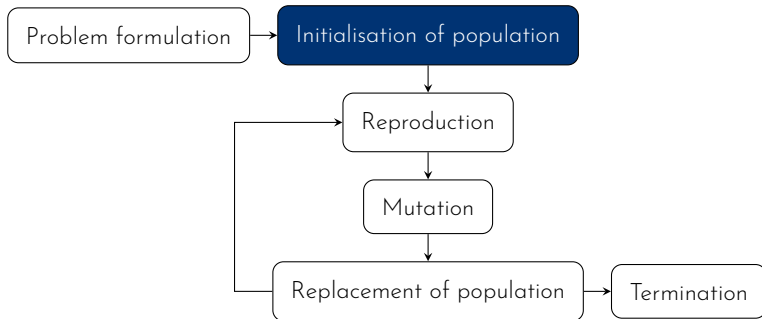
Genetic algorithm is an algorithm to solve an optimisation problem in which we are

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- ▶ to maximise the fitness of the chromosome

Genetic algorithm



Genetic algorithm



Initialisation of population

An initial population is generated with n number of chromosomes with random value assigned to the genes.

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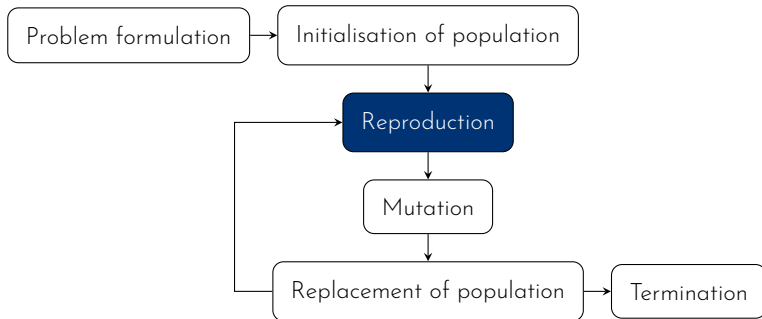
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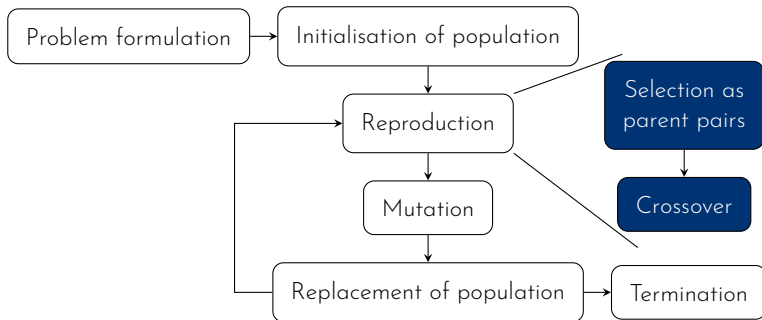
- ▶ **random value**

to make sure that the initial population uniformly represented the entire search space.

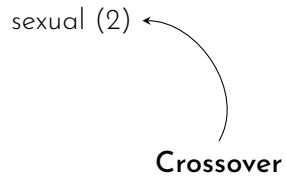
Genetic algorithm



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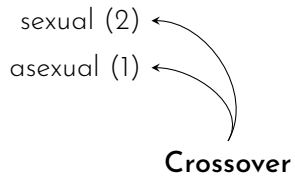


Crossover



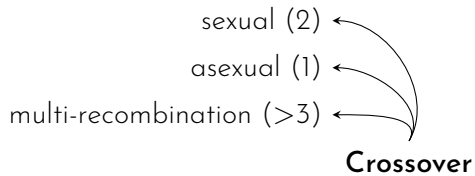
Reproduction

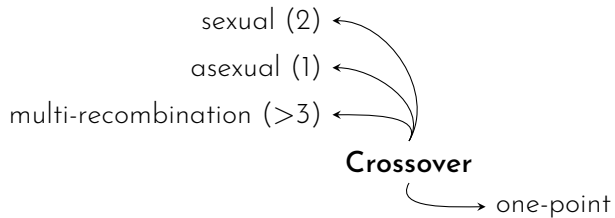
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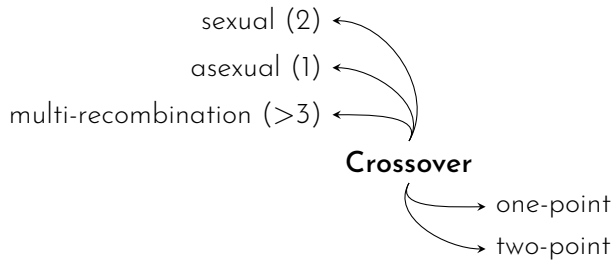


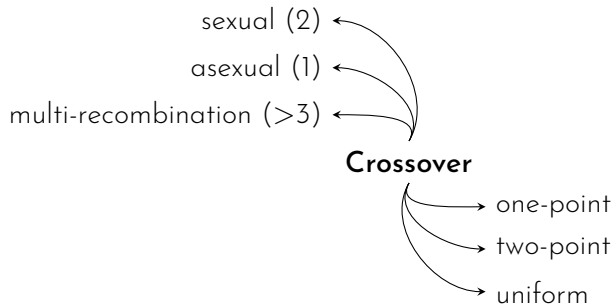
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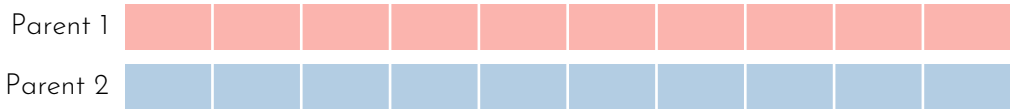




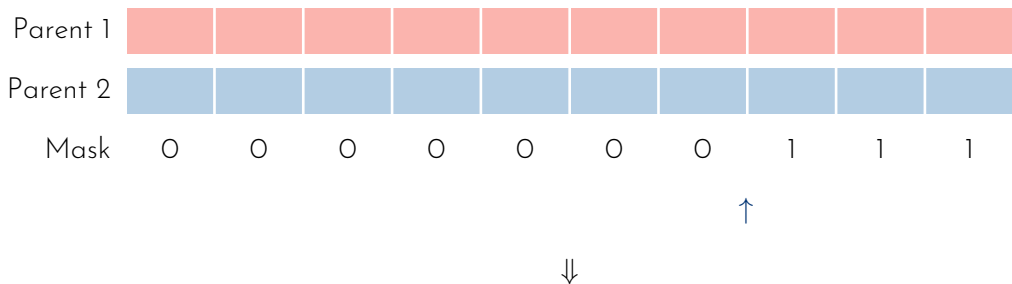




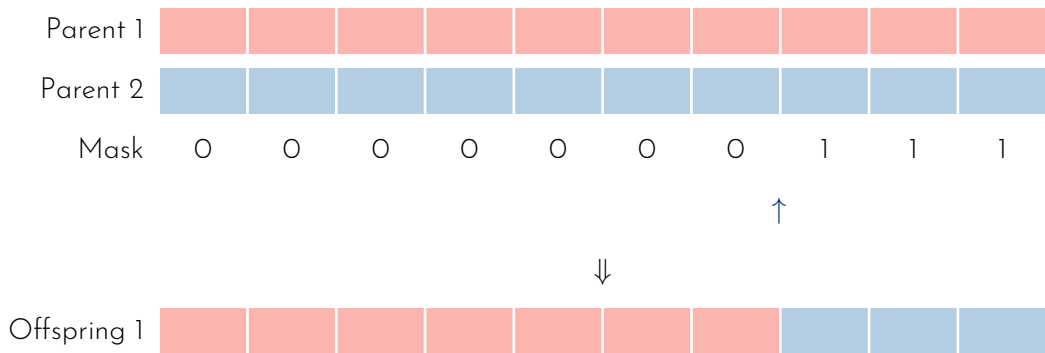
One-point crossover



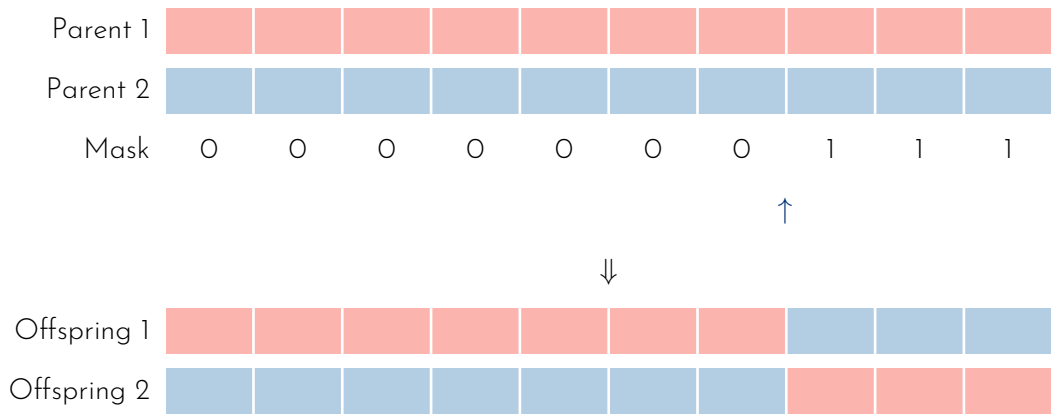
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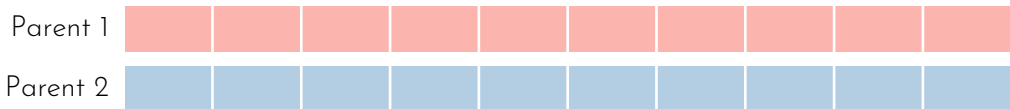
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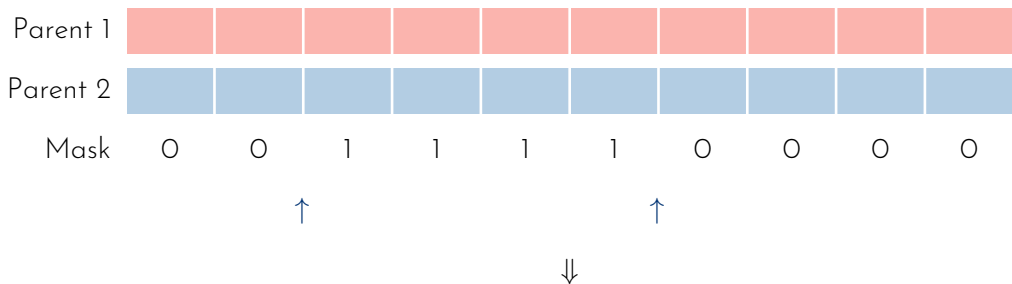
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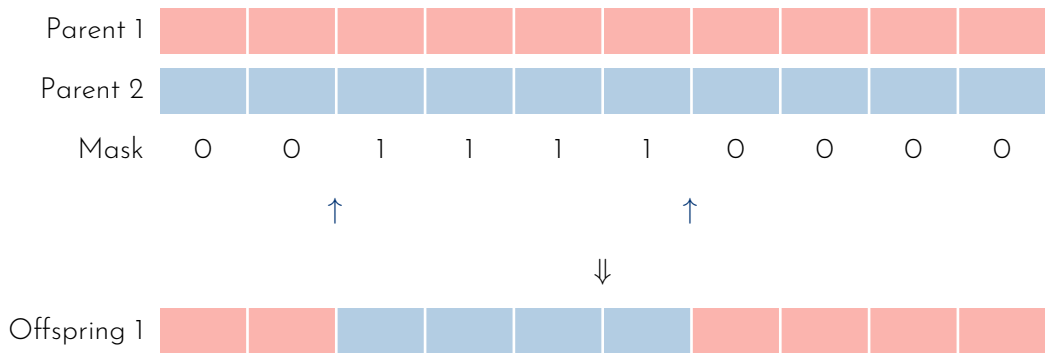
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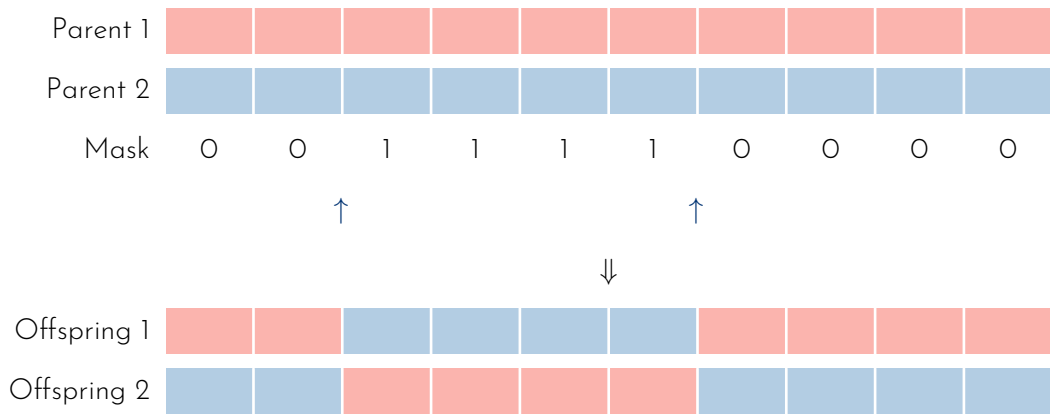
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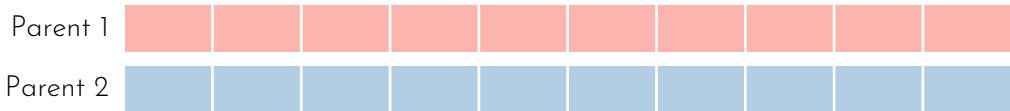
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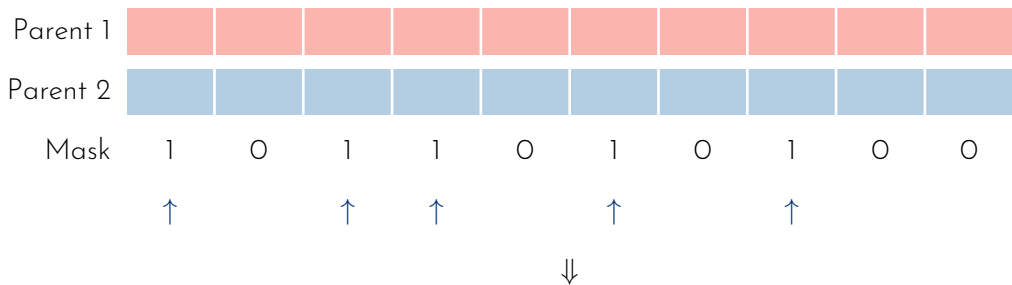
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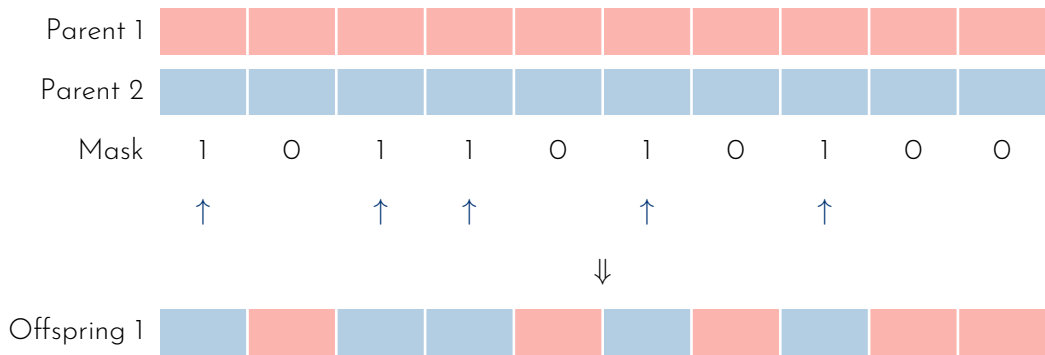
Uniform crossover



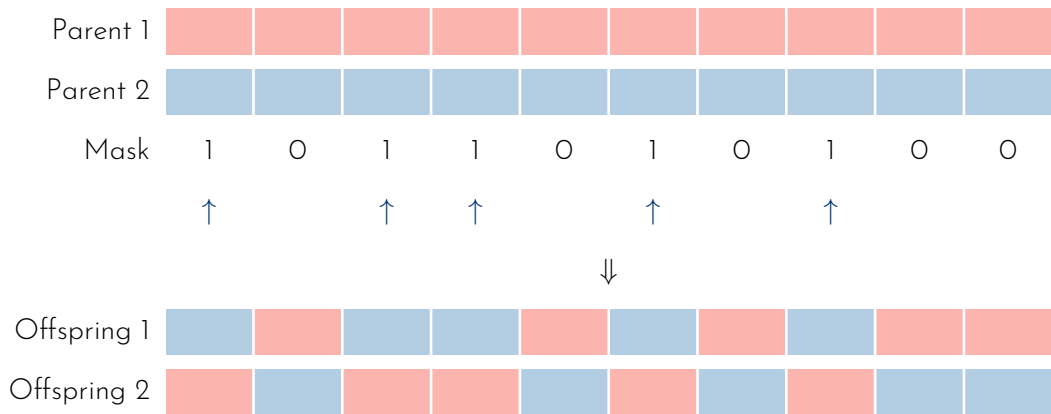
Uniform crossover



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How about asexual or multi-recombination?

Asexual crossover

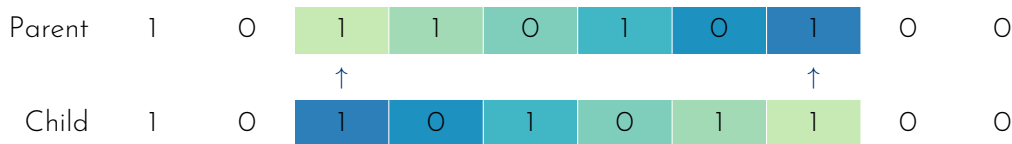
inversion – switching end to end of a portion of the parent



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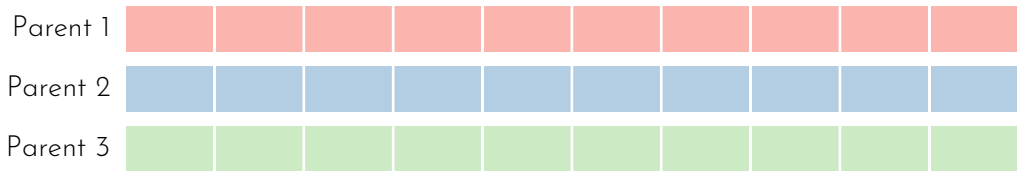
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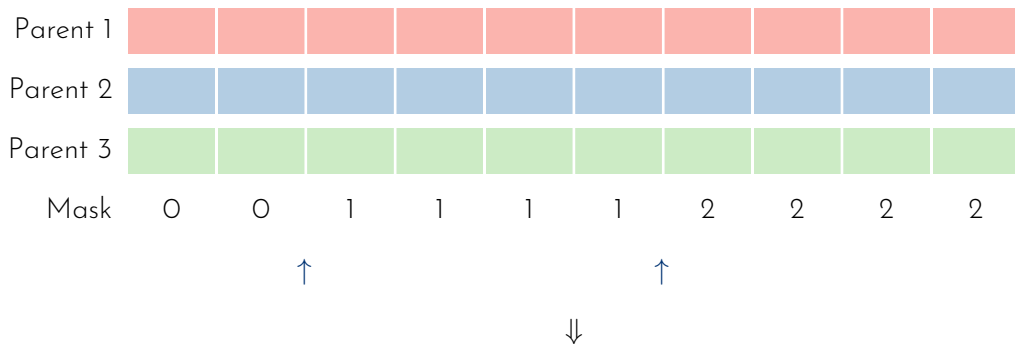
n -point crossover - with n parents, $n - 1$ crossover points are selected

Multi-recombination crossover

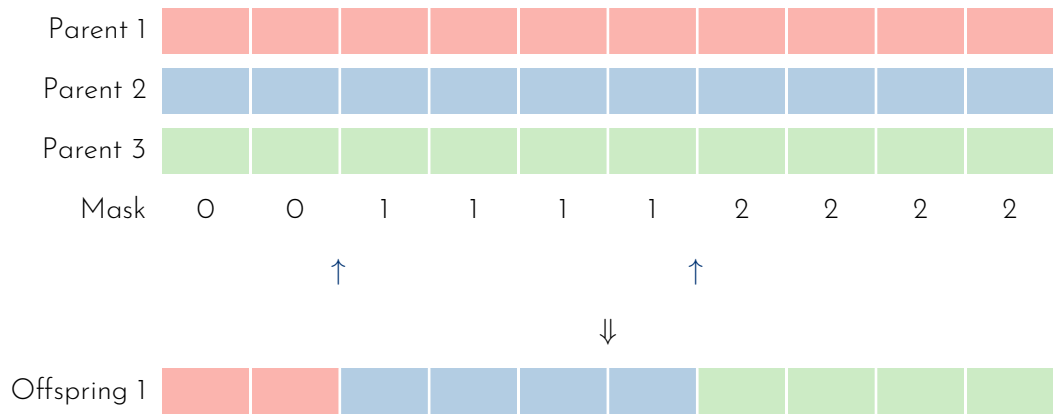
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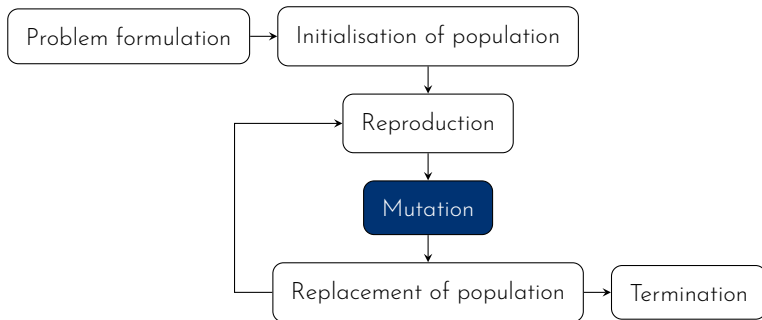
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Genetic algorithm



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- ▶ **uniform** (random) mutation and
- ▶ **inorder** mutation.

All bits have the equal mutation probability p_m .

Uniform mutation

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Before mutation	0	1	1	0	1	1	1	0	0	0
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After mutation	0	1	1	1	0	1	1	0	0	1

Two mutation points are randomly selected. Uniform/random mutation is performed within these points.

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Before
mutation

0

1

1

0

1

1

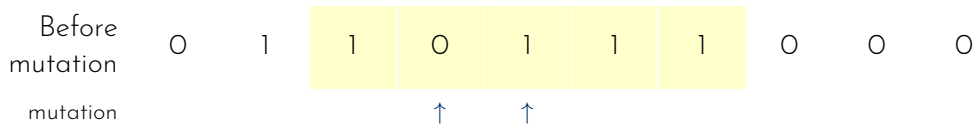
1

0

0

0

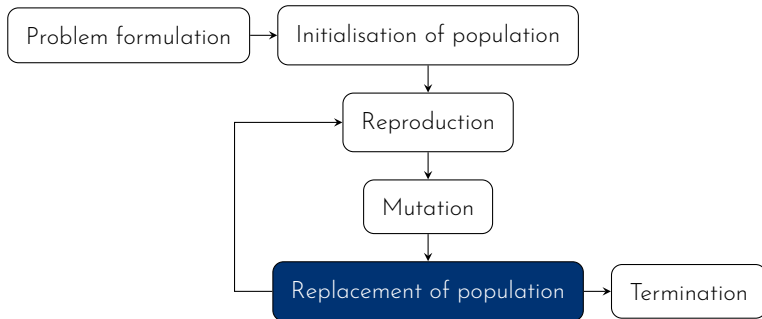
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Genetic algorithm



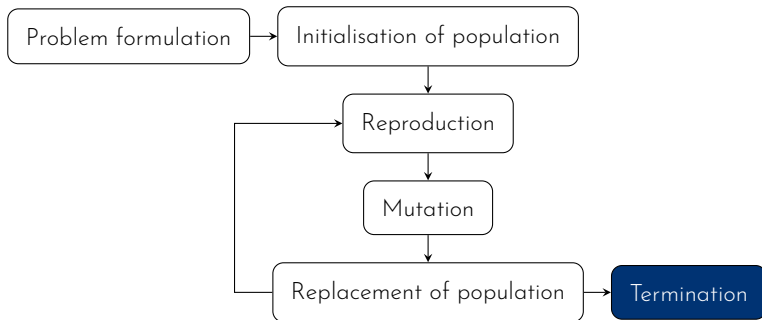
Replacement of Population

1. All children are selected; all parents are discarded; All children are used as the current generation to replace the population.

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2. Children and parents are selected based on fitness.

Genetic algorithm



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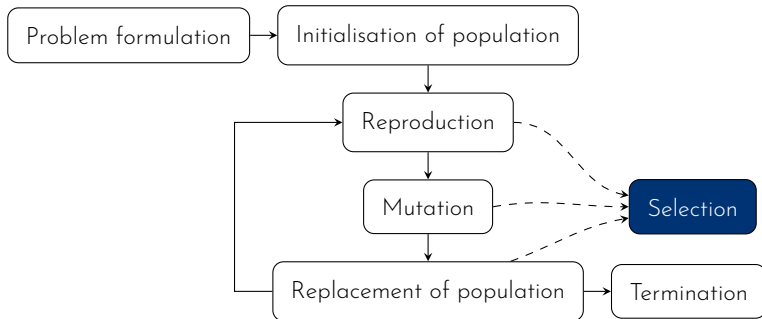
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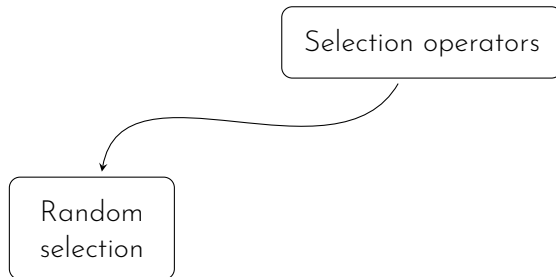
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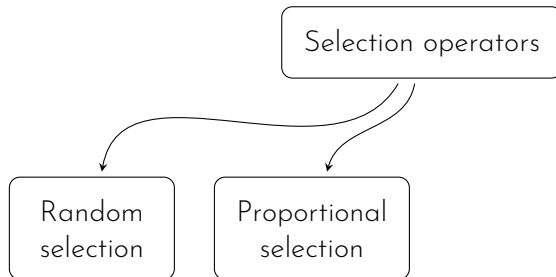
- ▶ termination due to no improvement is observed over a number of consecutive generations.
- ▶ termination due to no change in population.
- ▶ termination due to finding an acceptable solution.
- ▶ termination due to exceeding a maximum number of iterations.

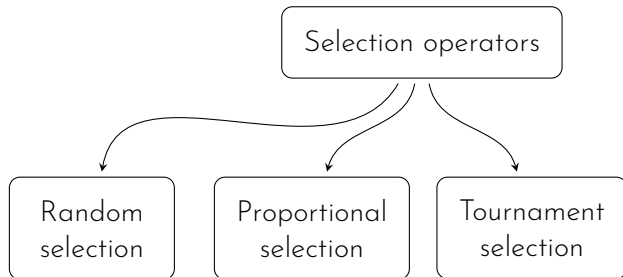
Genetic algorithm

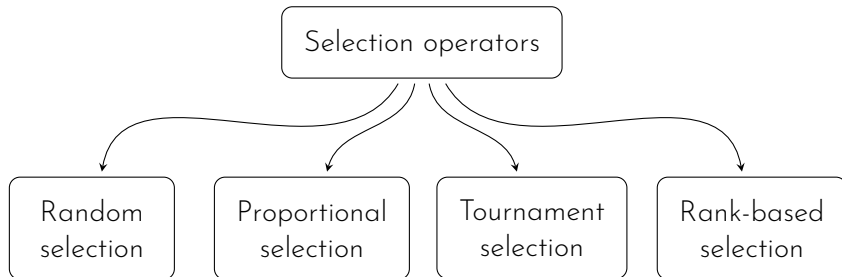


Selection operators









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- ▶ no fitness information is involved

- uses the fitness of each individual to calculate the probability that the individual is selected

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$$P(x_i) = \frac{F(x_i)}{\sum_{j=1}^{n_s} F(x_j)}$$

where,

$P(x)$ is the probability that x is selected

$F(x)$ is the scaled fitness of x

(to have positive value of the fitness)

n_s is the total number of individuals in the population

- ▶ selects a group n_{ts} of individuals randomly from the population and pick the best individual (individual with highest fitness)

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- ▶ n_{ts} affects how much the best individual dominates the selection result

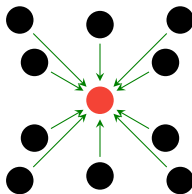
- individuals are ranked according to their fitness (in decreasing order)

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- ▶ the probability of selecting an individual is calculated based on its rank (instead of its actual fitness value)

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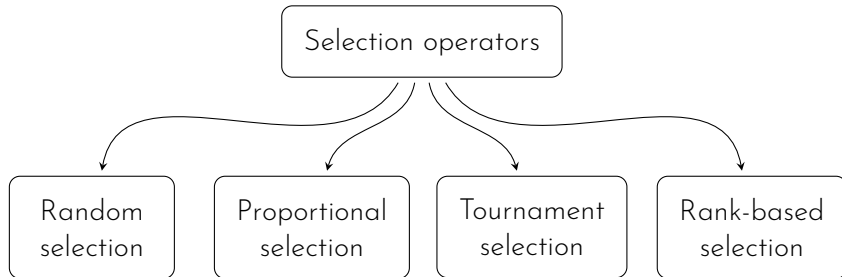
Selective pressure is defined as the speed at which the best solution will occupy the entire population by repeated application of the selection operator alone.



High selective pressure decreases diversity (thus limiting the exploration abilities of the population), may lead to premature convergence to suboptimal solutions.

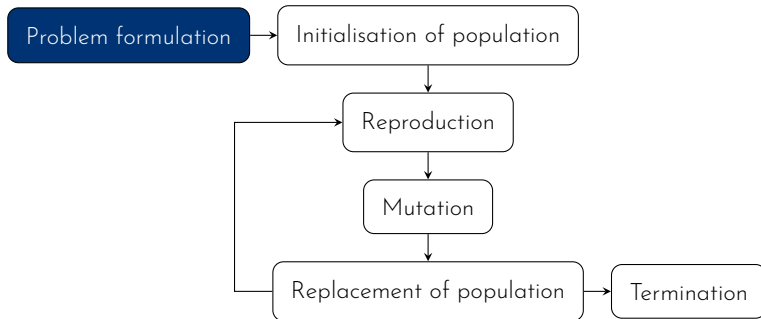
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Low selective pressure increases the time to convergence.



Compare the selective pressure of the different selection operators

Genetic algorithm



Problem formulation

- ▶ Encoding
- ▶ Fitness function

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genetic algorithm is an algorithm to solve an optimisation problem in which we are optimising the values of the chromosome to maximise the fitness of the chromosome.

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An individual chromosome is a potential solution for the optimisation problem. The variables to be optimised are genes in the chromosome.

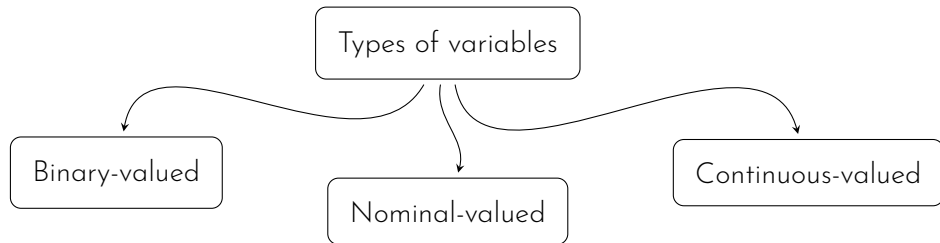
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The classic representation scheme for chromosomes are binary vectors with fixed length.

0	1	1	0	1	0
---	---	---	---	---	---

A chromosome can be used to represent different types of variables.



Binary-valued variables

- ▶ boolean data
- ▶ each variable occupies only 1 bit

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Example,

- ▶ one binary-valued variable: 0
- ▶ two binary-valued variables: 0 0
- ▶ three binary-valued variables: 0 0 0

Nominal-valued variables

- ▶ categorical data
- ▶ one-hot encoding

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- ▶ categorical data
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Example,

- ▶ a variable containing the information of weather
- ▶ set of possible values: {hot, windy, cloudy, rain, thunderstorm}
- ▶ encoded as a chromosome:

0	1	0	0	0
---	---	---	---	---

A nominal-valued variable with 2 possible states can be represented as a 1-bit binary string.

Is this statement true or false? why?

Continuous-valued variables

- ▶ mapped to a binary string using **binary coding**
 1. defining a finite range for the continuous value, $[x_{\min}, x_{\max}]$,

Continuous-valued variables

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For example, taking the range of 10 to 100, what is the 4-bit binary string representation of the value 16? What is the 5-bit binary string representation? What has changed when the number of bits is changed?

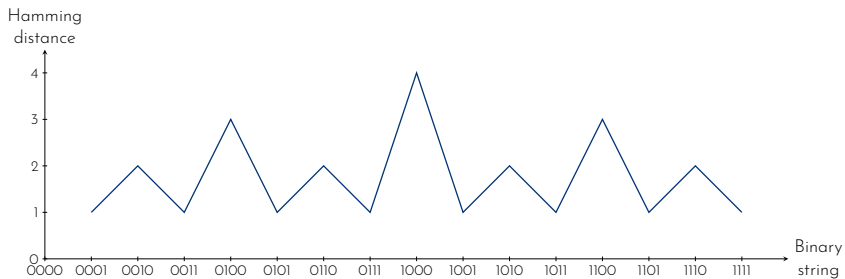
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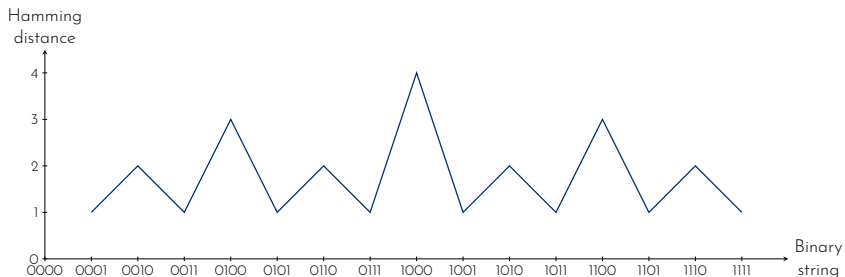
The maximum attainable accuracy of a binary string is

$$\frac{x_{\max} - x_{\min}}{2^{n_b} - 1}$$

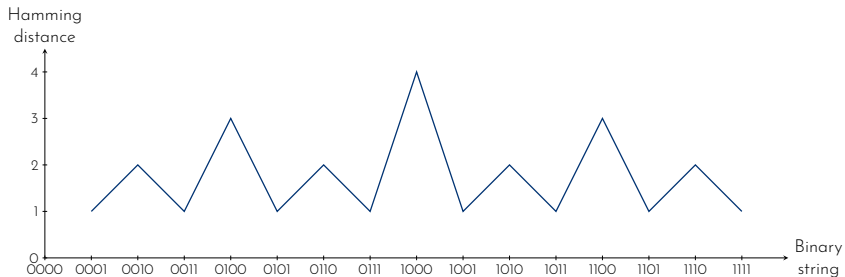
Binary coding causes the **hamming cliff**.



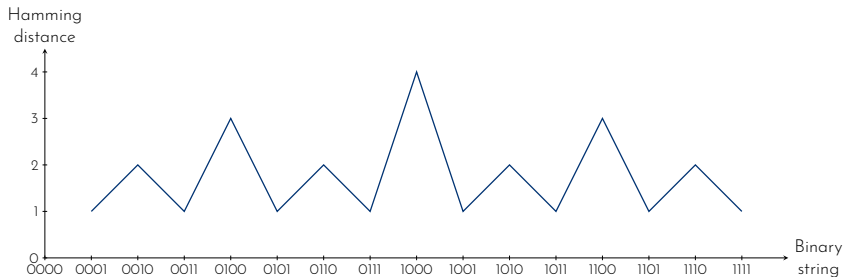
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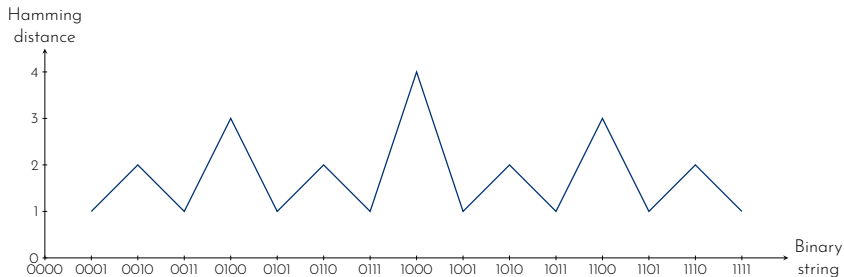
Hamming distance is the number of changing bits going from one binary string to another.



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Binary-to-Gray code conversion

(MSB) (LSB)

b_3 b_2 b_1 b_0

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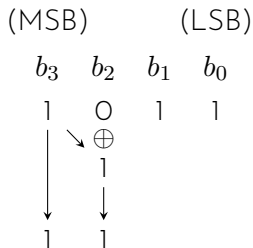
Binary-to-Gray code conversion

(MSB)			(LSB)
b_3	b_2	b_1	b_0
1	0	1	1
↓			
1			

Maintain the most significant bit (MSB)

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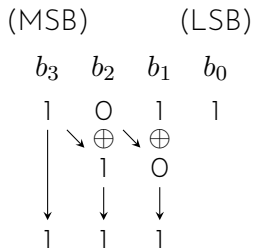
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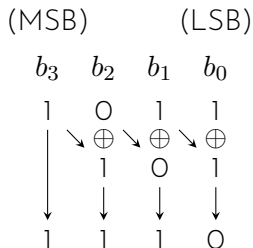
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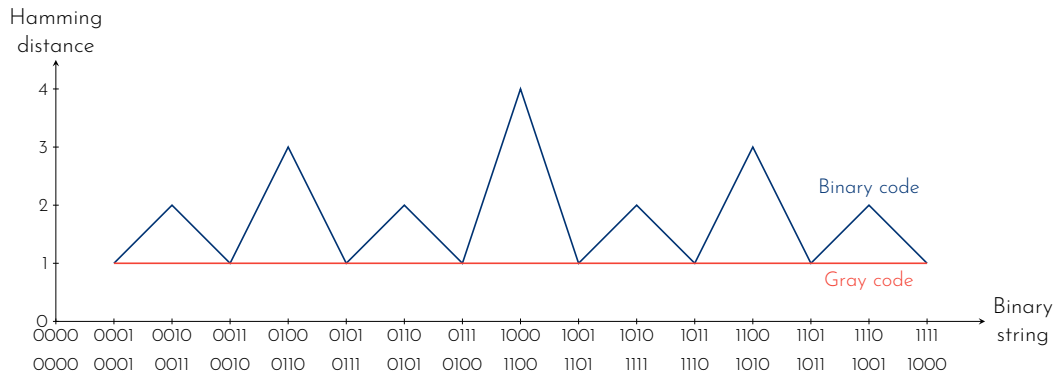
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Encoding

Gray code



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- ▶ Therefore a fitness function has to be carefully designed for meaningful comparison of fitness between different individuals.
- ▶ The fitness function takes input of a chromosome representation and produces a scalar value which represents the fitness of the chromosome.
- ▶ The higher the fitness value, the higher chance the chromosome will survive and be selected for reproduction.

Example 1

Problem: What is the maximum value of $f(x) = 15x - x^2$, assuming x can only be integers between 0 and 15?

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Integer	Binary code	Gray code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100

Example 1

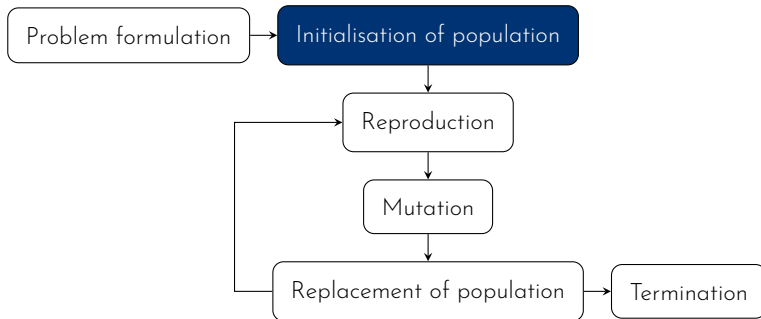
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6	0110	0101
7	0111	0100

Integer	Binary code	Gray code
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

Genetic algorithm



Example 1

Initialisation of population

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Let's define the population size N to be 10.

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$$v_3 = 0000_g = 0$$

$$v_4 = 1010_g = 12$$

$$v_5 = 0010_g = 3$$

$$v_6 = 1100_g = 8$$

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Can two individuals have the same value?

Example 1

Formulation of fitness function

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Since we are solving the problem of finding the maximum value of $f(x) = 15x - x^2$, we can use $f(x)$ as our fitness function. This is because, the larger the value of $f(x)$ for a potential solution, the closer the potential solution is towards the desired solution, and thus it should be of higher fitness.

Example 1

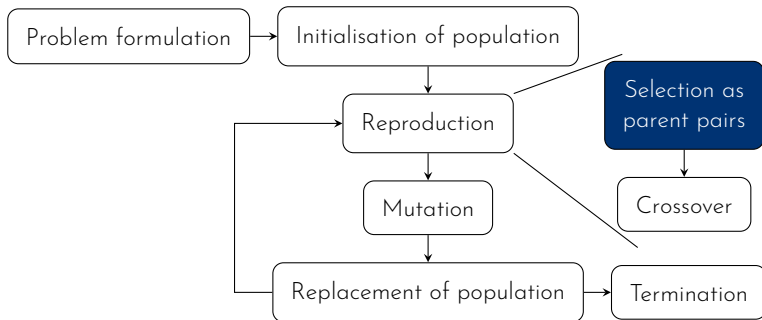
Calculating the fitness of each individuals using the fitness function $f(x) = 15x - x^2$

Example 1

Calculating the fitness of each individuals using the fitness function $f(x) = 15x - x^2$

Label	Binary string	Integer value	Fitness
v_1	0110	4	44
v_2	1111	10	50
v_3	0000	0	0
v_4	1010	12	36
v_5	0010	3	36
v_6	1100	8	56
v_7	1000	15	0
v_8	1011	13	26
v_9	0011	2	26
v_{10}	0001	1	14

Genetic algorithm



Example 1

Selection as parent pairs

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Using proportional selection, the probability of the selection of the individual is given by the fitness ratio $\frac{f(x_i)}{\sum_{\forall j} f(x_j)}$

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Label	Integer value	Fitness	Fitness ratio
v_1	4	44	0.1528
v_2	10	50	0.1736
v_3	0	0	0
v_4	12	36	0.125
v_5	3	36	0.125
v_6	8	56	0.1944
v_7	15	0	0
v_8	13	26	0.0903
v_9	2	26	0.0903
v_{10}	1	14	0.0486

Example 1

Proportional selection is also known as the roulette wheel selection.

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Visually we can **arrange the individuals on a roulette wheel** according to their fitness ratios. Then the selection process is carried out by simulating the turning of the roulette wheel.

Example 1

Starting from the first individual v_1 , the cumulative fitness ratio for each individual is calculated.

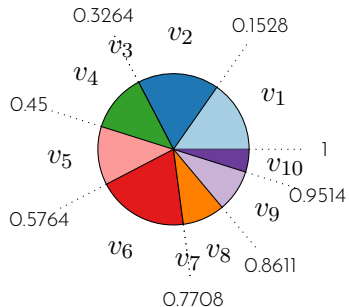
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Label	Fitness ratio	Cumulative fitness ratio
v_1	0.1528	0.1528
v_2	0.1736	0.3264
v_3	0	0.3264
v_4	0.125	0.4514
v_5	0.125	0.5764
v_6	0.1944	0.7708
v_7	0	0.7708
v_8	0.0903	0.8611
v_9	0.0903	0.9514
v_{10}	0.0486	1.0

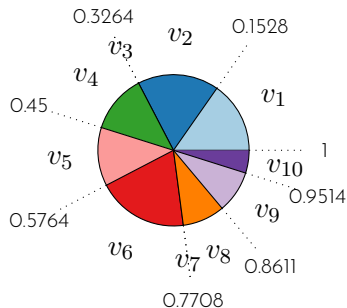
Example 1

Roulette wheel with all individuals



Example 1

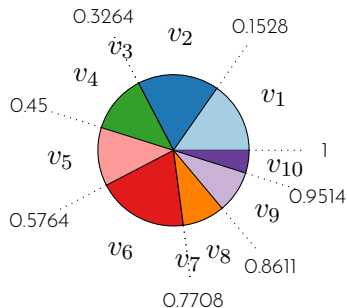
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To simulate selecting an individual by turning the roulette wheel,

Example 1

Roulette wheel with all individuals

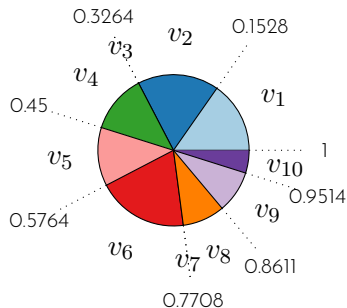


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1. select a random number between 0 and 1, and

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Roulette wheel with all individuals



To simulate selecting an individual by turning the roulette wheel,

1. select a random number between 0 and 1, and
2. select the individual within whose area the random number lies.

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Label	Parent A	Parent B
p_1	0.9189	0.1049
p_2	0.3305	0.0868
p_3	0.8211	0.9057
p_4	0.5780	0.9635
p_5	0.2476	0.8286

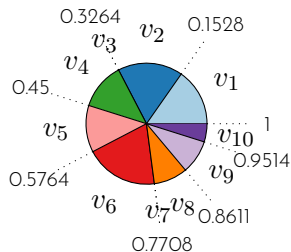
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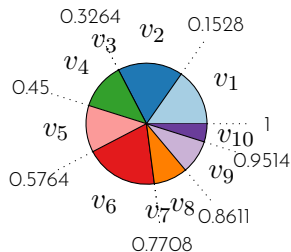
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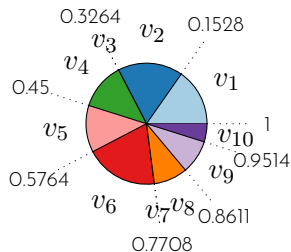
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p_1	0.9189	v_9	0.1049	
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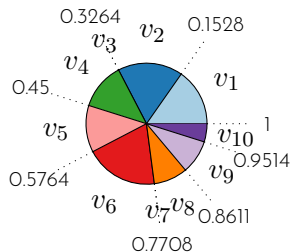
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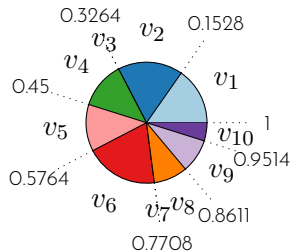
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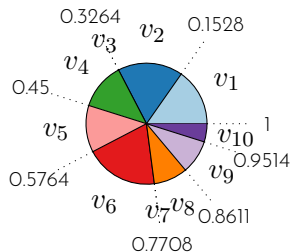
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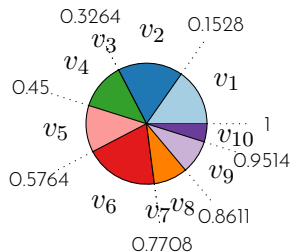
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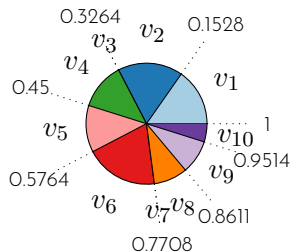
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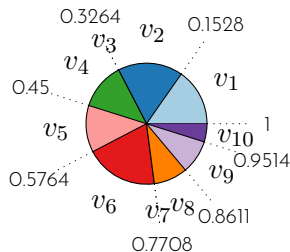
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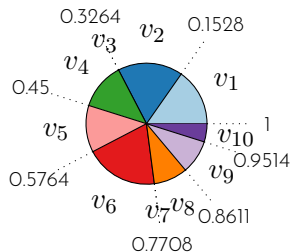
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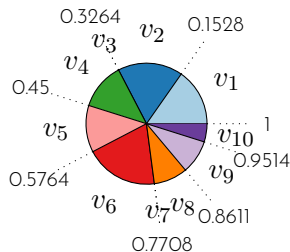
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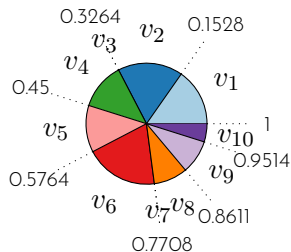
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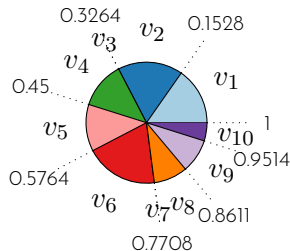


1. Do we allow one individual being selected in different parent pairs?

Example 1

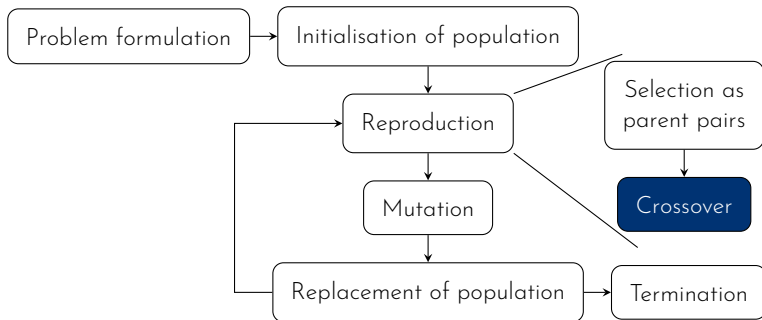
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1. Do we allow one individual being selected in different parent pairs?
2. Do we allow one parent pair consists of the same individual twice?

Genetic algorithm



Example 1

The next question is, how do we perform the crossover? How do we decide the crossover point?

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For each pair of parents, we generate a random integer number between 0 and $n - 2$ where n is the length of an individual, and 0 denotes the position of the left-most bit.

Example 1

The next question is, how do we perform the crossover? How do we decide the crossover point?

For each pair of parents, we generate a random integer number between 0 and $n - 2$ where n is the length of an individual, and 0 denotes the position of the left-most bit.

The crossover point is the point after the bit denoted by the random integer.

Example 1

Label		Parent A		Parent B		Random integer	Child 1	Child 2
p_1	v_9	0 0 1 1	v_1	0 1 1 0		2		
p_2	v_4	1 0 1 0	v_1	0 1 1 0		0		
p_3	v_8	1 0 1 1	v_9	0 0 1 1		2		
p_4	v_6	1 1 0 0	v_{10}	0 0 0 1		1		
p_5	v_2	1 1 1 1	v_8	1 0 1 1		1		

Example 1

Label	Parent A				Parent B				Random integer	Child 1	Child 2
p_1	v_9	0	0	1-1	v_1	0	1	1-0	2	0010	0111
p_2	v_4	1	0	1	0	v_1	0	1	1	0	
p_3	v_8	1	0	1	1	v_9	0	0	1	1	
p_4	v_6	1	1	0	0	v_{10}	0	0	0	1	
p_5	v_2	1	1	1	1	v_8	1	0	1	1	

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p_2	v_4	1-0 1 0	v_1	0-1 1 0		0	1110	0010
p_3	v_8	1 0 1 1	v_9	0 0 1 1		2		
p_4	v_6	1 1 0 0	v_{10}	0 0 0 1		1		
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p_3	v_8	1 0 1-1	v_9	0 0 1-1		2	1011	0011
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p_5	v_2	1 1 1 1	v_8	1 0 1 1		1		

Example 1

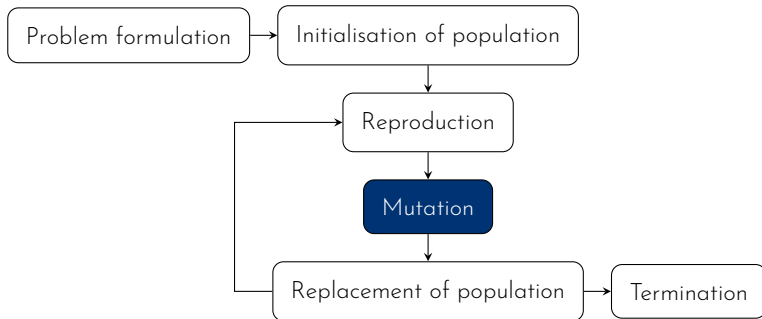
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p_4	v_6	1 1-0 0	v_{10}	0 0-0 1		1	1101	0000
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Example 1

Let's label the children

c_1	0010	c_6	0111
c_2	1110	c_7	0010
c_3	1011	c_8	0011
c_4	1101	c_9	0000
c_5	1111	c_{10}	1011

Genetic algorithm



Example 1

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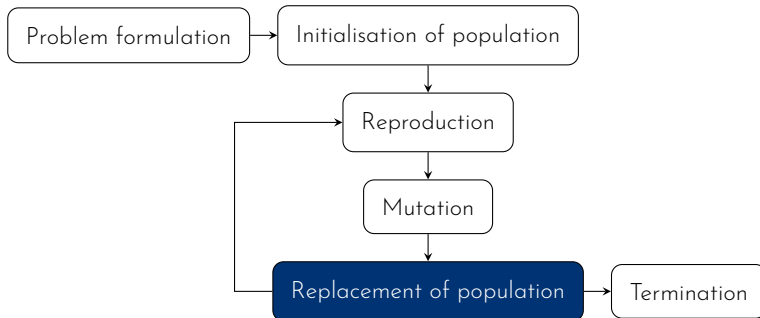
$$c_2 \quad \underline{111}0 \rightarrow 1\underline{1}00$$

Example 1

The final versions of the children produced are

Child	Gray code	Decimal value	Fitness
c_1	0010_g	3	36
c_2	1100_g	8	56
c_3	1011_g	13	26
c_4	1101_g	9	54
c_5	1111_g	10	50
c_6	0111_g	5	50
c_7	0010_g	3	36
c_8	0011_g	2	26
c_9	0000_g	0	0
c_{10}	1011_g	13	26

Genetic algorithm



Example 1

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Repeat the process of crossover and mutation until one of the termination criteria is fulfilled.

Example 1

The average fitness of the original generation:

$$\frac{44 + 50 + 0 + 36 + 36 + 56 + 0 + 26 + 26 + 14}{10} = 28.8$$

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The new generation has a higher average fitness compare to the original generation.
The population has, overall, become better.

Can we be sure that, based on the mechanism of genetic algorithm and the synergy of different processes in genetic algorithm, the population is **evolving instead of degenerating**?

Schemata theorem shows that the population will contain more individuals with better fitness after multiple generations.

Taking an example of 10 chromosomes with the following fitness values (higher better)

Label, c_i	Chromosome	Fitness, f_i
c_0	0000	4
c_1	0001	10
c_2	0011	18
c_3	0010	24
c_4	0110	28
c_5	0111	30
c_6	0101	30
c_7	0100	28
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where * denotes “don’t care”.

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The generalised representation is called a schema (plural: schemata)

Schemata theorem

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$$P(c_5 \text{ or } c_6) = \frac{f_5 + f_6}{\sum_{i=0}^9 f_i} = \frac{2 \times f_{5/6}}{\sum_{i=0}^9 f_i}$$

where $f_{5/6}$ is the average fitness of chromosomes c_5 and c_6 .

Assuming the generation goes through selections of 10 chromosomes before the next generation,

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where f_{avg} is the average fitness of the whole population

$$m_{5/6}(t+1) = \frac{2 \times f_{5/6}}{f_{\text{avg}}}$$

Will $m_{5/6}(t+1) > 2$ or $m_{5/6}(t+1) < 2$?

Let's generalise the equation,

$$m_{5/6}(t+1) = \frac{2 \times f_{5/6}}{f_{\text{avg}}}$$

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- ▶ 2 is the number of chromosomes with schema $O1^*1$ at current generation, $m_H(t)$

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- What happens to schemata with fitnesses lower than the average fitness?

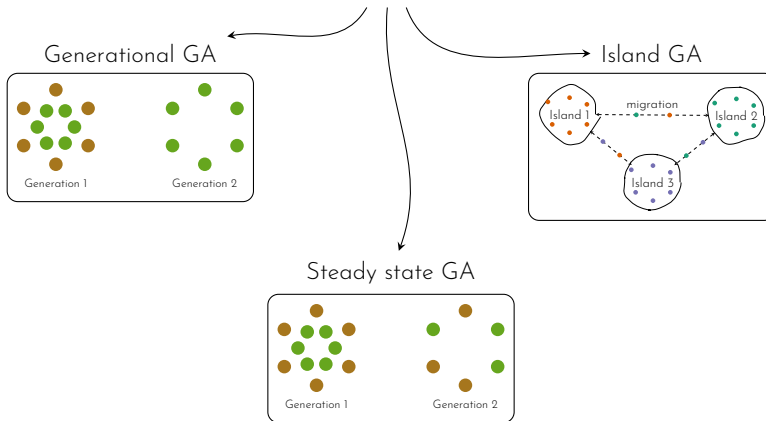
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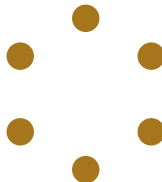
- ▶ What happens to schemata with fitnesses lower than the average fitness?
- ▶ What happens to schemata with fitnesses higher than the average fitness?
- ▶ How would the average fitness of the next generation compare to the previous generation?

Variant on replacement strategy,
i.e. how the offsprings replace the parents.



Generational genetic algorithm (GGA)

- ▶ all parents are replaced by their offsprings in the next generation



Generation 1

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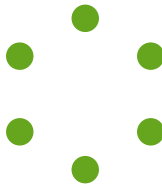
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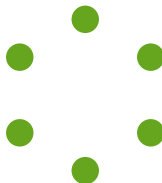
Generation 2

Generational genetic algorithm (GGA)

- ▶ all parents are replaced by their offsprings in the next generation
- ▶ no overlapping between the current and the new population. The overlapping between current and new population is referred to as **generation gap**.



Generation 1



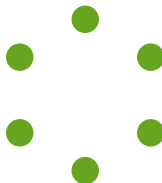
Generation 2

Generational genetic algorithm (GGA)

- ▶ all parents are replaced by their offsprings in the next generation
- ▶ no overlapping between the current and the new population. The overlapping between current and new population is referred to as **generation gap**. GGAs have zero generation gap.



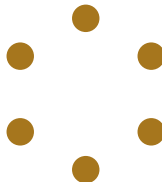
Generation 1



Generation 2

Steady state genetic algorithm (SSGA)

- ▶ once an offspring is created, a decision to whether the parent or the offspring survives to the next generation



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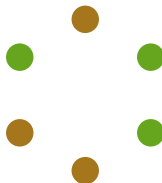
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Generation 1



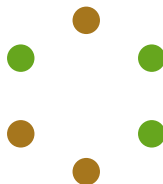
Generation 2

Steady state genetic algorithm (SSGA)

- ▶ once an offspring is created, a decision to whether the parent or the offspring survives to the next generation
- ▶ there is an overlap between the current and the new population. SSGAs have large generation gaps.



Generation 1



Generation 2

Examples of replacement strategies:

- replace worst – offspring replaces the worst individual

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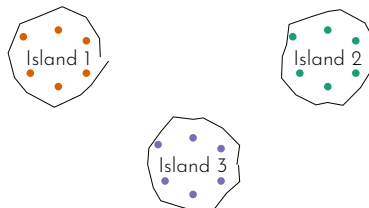
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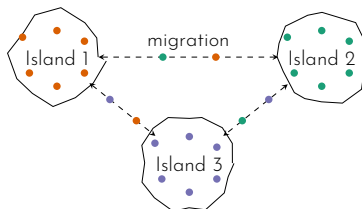
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- ▶ kill tournament - randomly select a group, offspring replace the worst in the group
- ▶ replace oldest - offspring replace the oldest individual
- ▶ parent-offspring competition - decide if a offspring replaces one of its parents

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- ▶ each population evolve within the island (selection, crossover, and mutation) independently of other islands



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- ▶ each population evolve within the island (selection, crossover, and mutation) independently of other islands
- ▶ migration happens when an individual from an island moves and replaces another individual in another island





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Thank you

CSC3034 Current Course Update

04 Genetic Algorithm - presentation

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current Lab 3 covers a complete GA workflow and continuous waypoint encoding.
- The Isaac Sim extension evolves collision-aware mobile-robot trajectories.
- A Matplotlib fallback demonstrates the optimization without simulator hardware.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.